

PORTABLE DMM Racing Forward, BENCHTOP DMM Holding On



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Since their inception in the 1960s, digital multimeters (DMMs) have undergone many changes. With features like auto-ranging, infrared (IR) and surface-temperature measurements integrated into a single device, the DMM is no longer a tool just for measuring circuit parameters. It has developed into a very versatile instrument with extreme safety features incorporated into it.

With demand for portable equipment, hand-held DMM is being used more than its benchtop brother. Certain security ratings and wireless additions have added to the safety of maintenance, repair and operations staff working with high-tension wires. Let us take a look at the availability, development and advances in these handy devices.

There is something in the air

Requirement for highly-accurate and precise results has lead to the emergence of certain added features besides regular measurement that help in fine-tuning the measurements. Arindam Majumdar, managing director, Fluke India, states, "One emerging trend in the global test and measurement (T&M)

equipment market is wireless connectivity of T&M equipment."

With addition of wireless features, we now have smart-DMMs capable of providing measurements to mobile applications. Keysight offers U1177A IR-to-Bluetooth adaptor, which can be connected to their DMM, in order to gain wireless capability. Users seem to have taken a certain liking to such devices. The ability to operate a DMM from afar without any harm to life seems to be the added motivation behind this change.

There has been a rapid increase in the manufacturing of such equipment. Such wireless DMMs employ Bluetooth Low Energy (BLE) in their operation, hence performing operations at the lowest power possible. An example of such wireless capability would be 3000 FC DMM from Fluke. With remote-logging and wireless capability, it can send measurement data to a designated smartphone. True-RMS measurements, 6000-count resolution, manual and automatic ranging provide a very wide set of measurements. Additionally, it provides higher 0.09 per cent basic accuracy, a digital display, up to three secondary measurements from remote modules and a bright white backlight.

Safer measurements

Wireless-enabled test equipment have a safety angle. These let the technician turn off the supply, place the test equipment while wearing insulated personal protective equipment and turn it back on from a safe distance, hence protecting self and reading test parameters on a smartphone or computer system. There are other safety measures to be found in modern DMMs.

Most multimeters include a fuse or two fuses, which prevent damage to the multimeter from a current overload on the highest current range. A common error when

Fluke 28 multimeter
input protection

